ModelIt!

Student Activity Pack

Biomimicry Design Lab

Engineering Solutions Inspired by Nature

Grade NE

NGSS MS-LS4-4, MS-ETS1-1, MS-ETS1-2, MS-ETS1-3, MS-ETS1-4

Structure and Function

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **In Level 1, you discovered that tail shape affects swimming speed. What OTHER animal features might engineers want to copy?**

2. **What makes a robot 'lifelike'? List as many factors as you can.**

3. **Why would a wildlife researcher want a robot that looks like a real animal?**

4. **What is a 'tradeoff' in engineering? Give an example from everyday life.**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Animal Feature Selected | Material Choice | Power Source | Movement Accuracy | Energy Use | Durability | Detection Risk

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: High Fidelity

Lifelike materials, moderate power

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Budget Build

Cheap materials, small power

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Power Boost

Maximum power source

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

ModelIt!

Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Cost — Total expense to build the design — more lifelike materials and larger power sources cost more

Environmental Resistance — How well the design handles rain, mud, extreme temperatures, and UV exposure

Noise Level — How much sound the design produces during operation — quieter is harder to detect

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Can engineers build a robot so lifelike that real animals cannot tell it apart from one of their own? Use evidence from your simulation to support your answers.

1. **Why does the fundamental feature choice matter more than materials or power?**

2. **Describe a tradeoff in your model where improving one output made another worse.**

3. **How is iterative design similar to evolution by natural selection?**

4. **What would it take to build a robot that is truly indistinguishable from a real animal?**

5. **How did running multiple scenarios help you understand the design space?**

8 STEM Challenge: Animal Robot Sketchbook

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YOUR ENGINEERING MISSION

Choose a real animal, research three of its key adaptations, then design a robot that borrows those features. Create a detailed engineering sketchbook with design rationale based on model data.

THE SCENARIO

A wildlife research organization needs robots that can observe animals without disturbing them. Your team will design an animal-mimicking robot, using your model to make and justify every design decision.

Engineering Design Thinking

1. WHAT ARE THE THREE MOST IMPORTANT ADAPTATIONS OF YOUR CHOSEN ANIMAL?

2. WHAT MATERIAL WILL YOU USE FOR EACH ADAPTATION AND WHY?

3. HOW MUCH POWER DOES YOUR DESIGN NEED — AND HOW WILL YOU PROVIDE IT?

4. WHAT IS THE BIGGEST WEAKNESS OF YOUR DESIGN, AND COULD A FUTURE ITERATION FIX IT?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Robotics Engineers specializing in biomimicry design robots inspired by nature for exploration, medical devices, search and rescue, and environmental monitoring. They earn $80,000–$140,000/year.